BALAJI CHIDAMBARAM

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SUMMARY

As an aspiring software engineer, my expertise lies in creating sophisticated data pipelines in an event-driven architecture, blending deep technical knowledge in AWS, Docker, SQL, and Kubernetes with a keen understanding of artificial intelligence, system architectures, and cloud technologies. My professional journey demonstrates a strong commitment to elevating data infrastructures and insights across diverse domains.

EDUCATION

University of Southern California — **Jan 2022-Dec 2023**
Master of Science in Computer Science — 3.81/4.0

Anna University — **Aug 2017-Jul 2021**
Bachelor of Engineering in Computer Science and Engineering — 8.97/10

SKILLS

- **Programming Languages:** Python, Scala, C and C++, Java, Javascript, Typescript, SQL, Swift, Golang, HTML5
- **Software Development & Libraries**: PyTorch, Tensorflow, Scikit-learn, Flask, NodeJS, Express.js, Streamlit, Keras, Scrapy, MySQL, PostgreSQL, MongoDB, GraphQL, Cassandra, Neo4j, ReactJS, Angular, SwiftUI, Tableau, Bootstrap, CSS
- **Tools, Platforms & Infrastructure:** Git, Apache Spark, PySpark, Airflow, Kubernetes, Docker, Jenkins, Linux, GCP, AWS
- **Domain Knowledge:** System Design, Data Engineering, Machine Learning, Deep Learning, NLP, Knowledge Graphs, Big Data

WORK EXPERIENCE

Marketing Analyst Intern - Keck Medicine of USC — **Sep 2023-Dec 2023**

- Took ownership of designing and deploying AWS-based data ingestion and ETL data pipelines, focusing on data quality and governance.
- Engineered a data infrastructure architecture in Adverity, integrating Salesforce marketing data with in-house volume data. This infrastructure enhancement amplified insights for marketing campaigns, ensuring data-driven decision-making.
- Developed a comprehensive analytics platform integrating Tableau, AWS, and Looker Studio to craft insightful dashboards. Achieved an increase of 54% in campaign impressions and a 32% uptick in engagement as a result of these dashboards.

Data Engineer Intern - Sayari Labs — **May 2023-Jul 2023**

- Designed and managed high-performance data pipelines as microservices using Scrapy, Kubernetes, and Airflow, transforming financial data and ensuring scalability and reliability. Maintained and updated data pipelines regularly, ensuring high-quality data by integrating over 10 data sources monthly.
- Enhanced data pipeline performance within a Spark cluster, standardizing transformation for improved software and product team data access.Developed end-to-end pipeline frameworks following industry best practices, focusing on efficient software solutions, test automation, CI/CD, and documentation.
- Learned to work on cross-functional projects and maintain code environments using agile methodologies, enhancing pipelines through iterative and rigorous testing and CI/CD automation via GitHub Actions.

Research Assistant - Information Sciences Institute — **Aug 2022-May 2023**

- Developed a knowledge graph algorithm with the KGTK toolkit, mapping complex relationships between entities and achieving a 78% accuracy rate in predicting information flow.
- Designed and implemented a chatbot capable of parsing GitHub repository README files into a knowledge graph and providing relevant answers and documentation to user queries.
- Enhanced user accessibility and understanding of GitHub projects by enhancing automation and data retrieval processes.

Cloud Architect Intern - HertzAI — **Aug 2021-Nov 2021**

- Streamlined development of cloud services with Docker automation techniques and CI/CD packaging with Jenkins for test automation, deployment, and release, resulting in accelerated production rollout within 2 months.
- Designed continuous improvement and deployment (CI/CD) pipelines with AWS CodePipeline, used AWS Elastic Container Registry for cloud hosting and service integration with API endpoints served using Flask, decreasing release time by 25%.

PROJECTS

- **Conversational AI with Retrieval Augmented Generation:** Utilized Llama 2 embeddings and Chroma to transform JSON error logs into an intuitive conversational AI-driven database. Developed robust language models using LangChain and Vicuna-13B LLM for Retrieval Augmented Generation. Enhanced natural language query processing and answer retrieval accuracy.
- **Automated Stock Market Streaming System with Spark and Kafka:** Developed a real-time stock market data streaming system in a distributed systems architecture using Kafka, with data structuring via PySpark Structured Streaming and storage in Cassandra. Automated deployment and orchestration were achieved using Docker, Airflow, and Kubernetes.